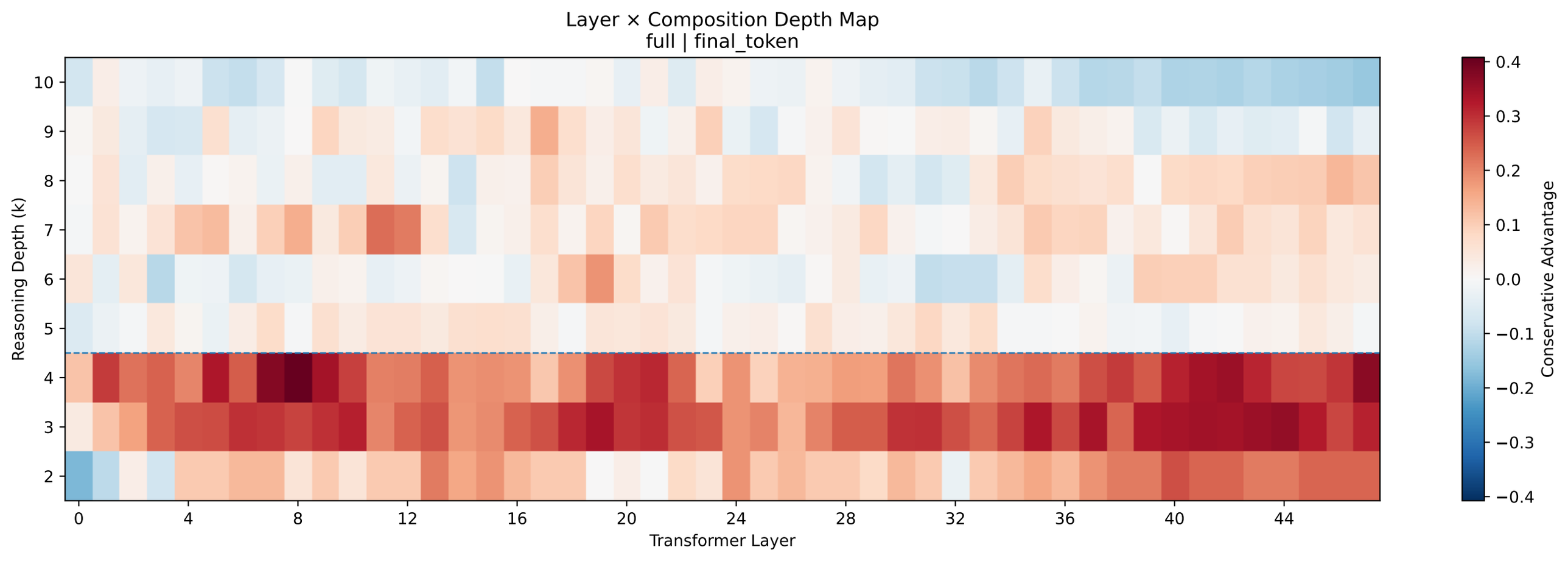
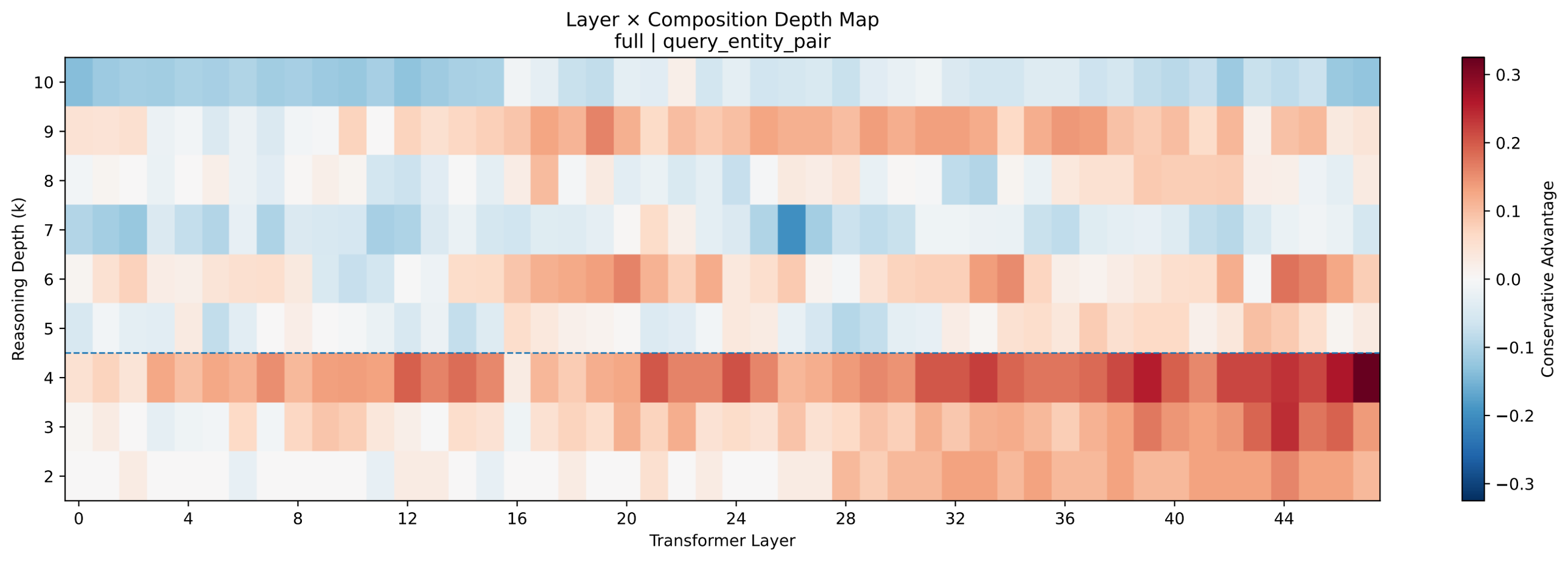
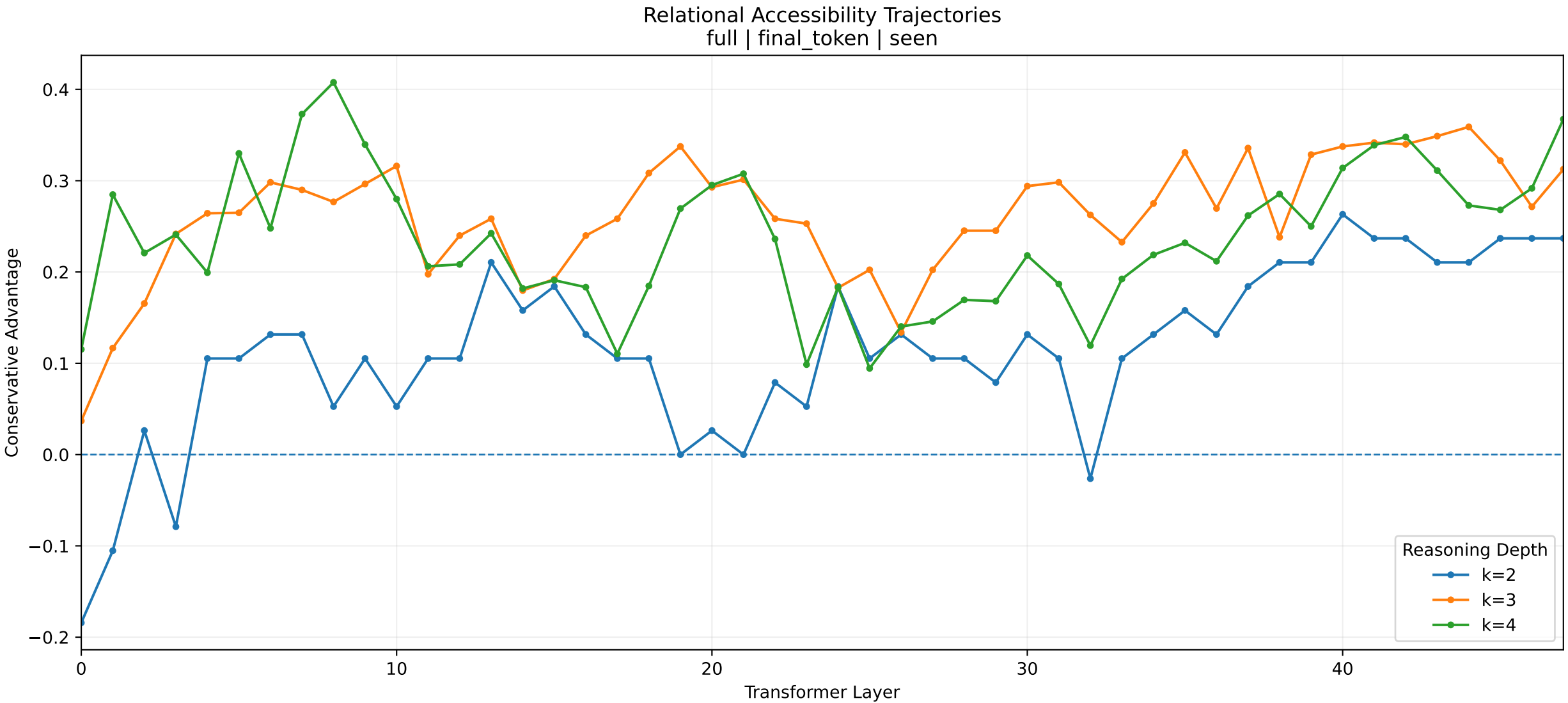
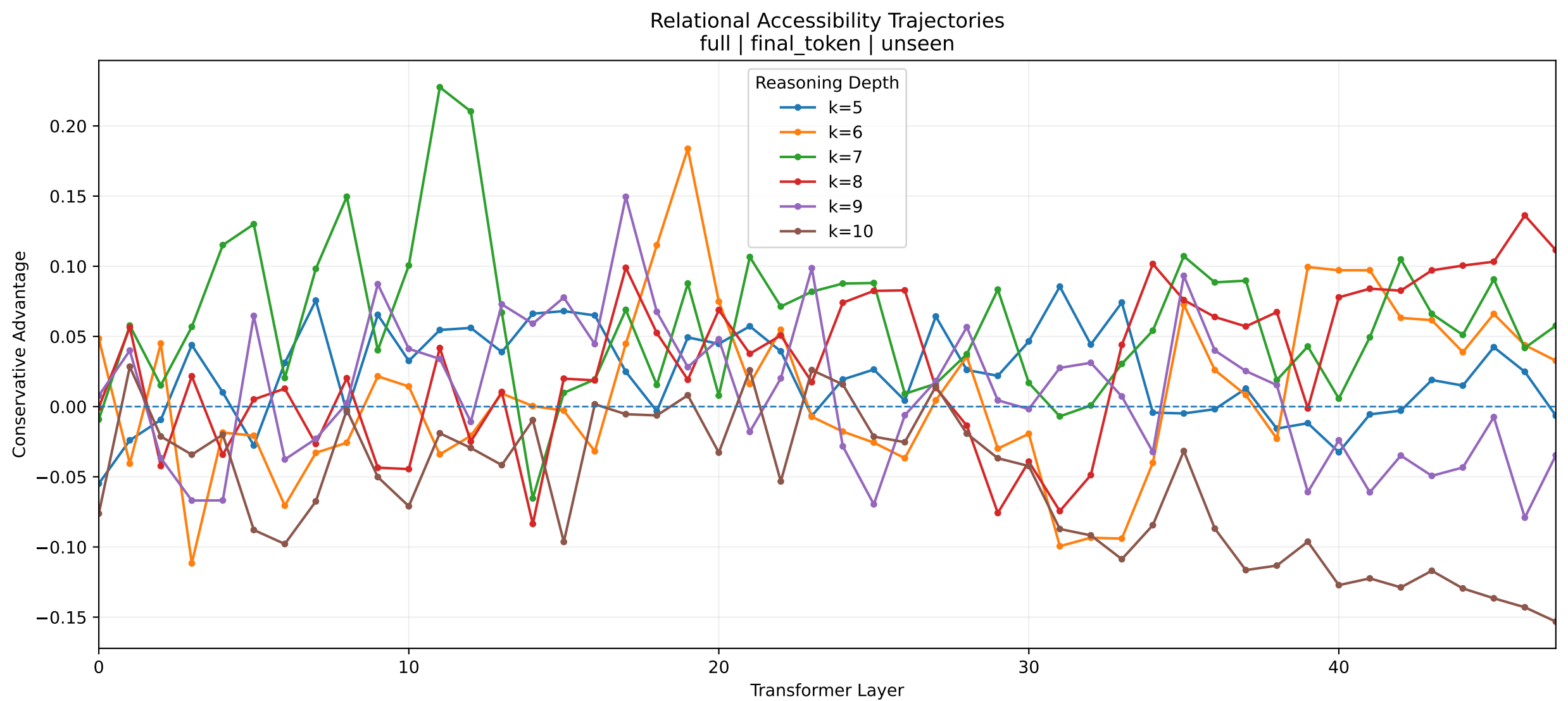
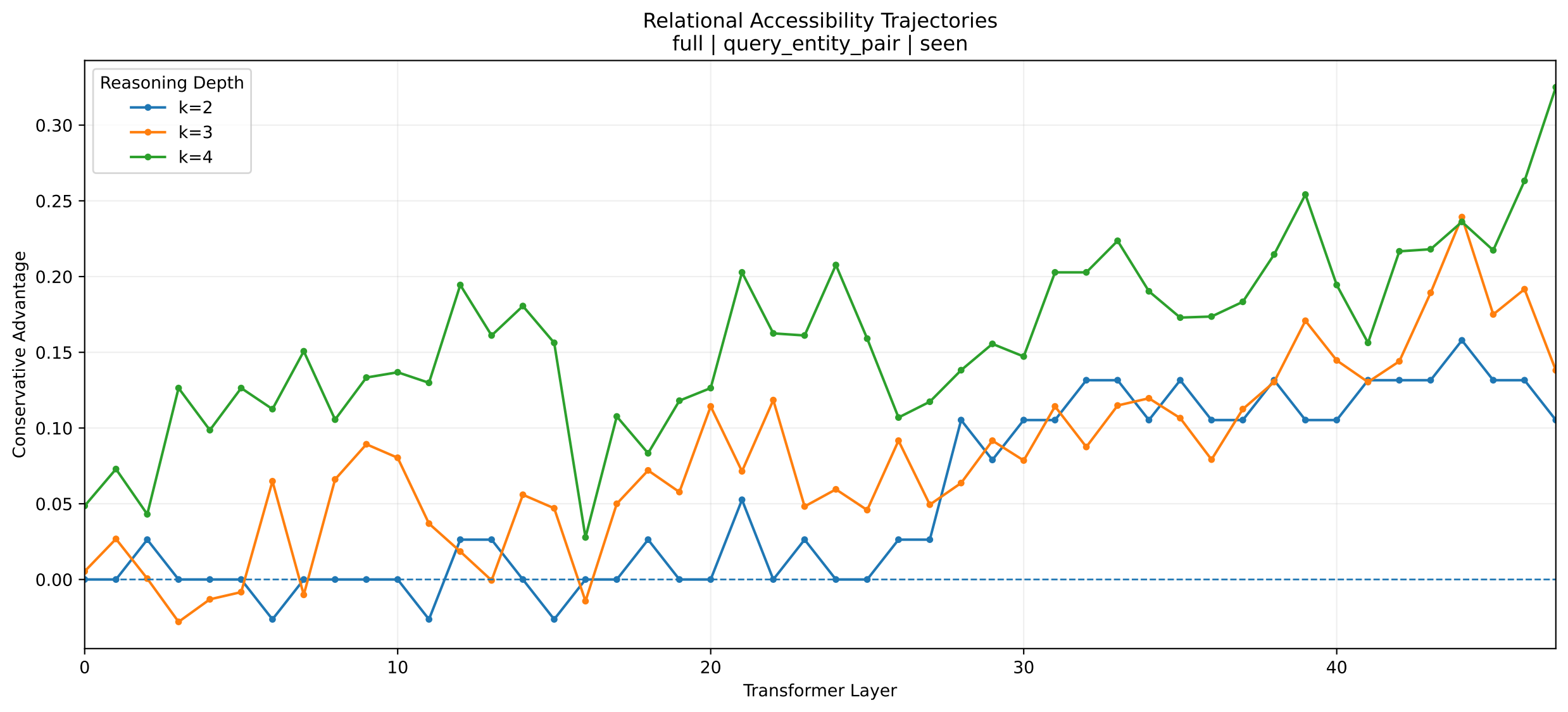

Layer × Composition Depth Map
full | final_token
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
Conservative Advantage
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4

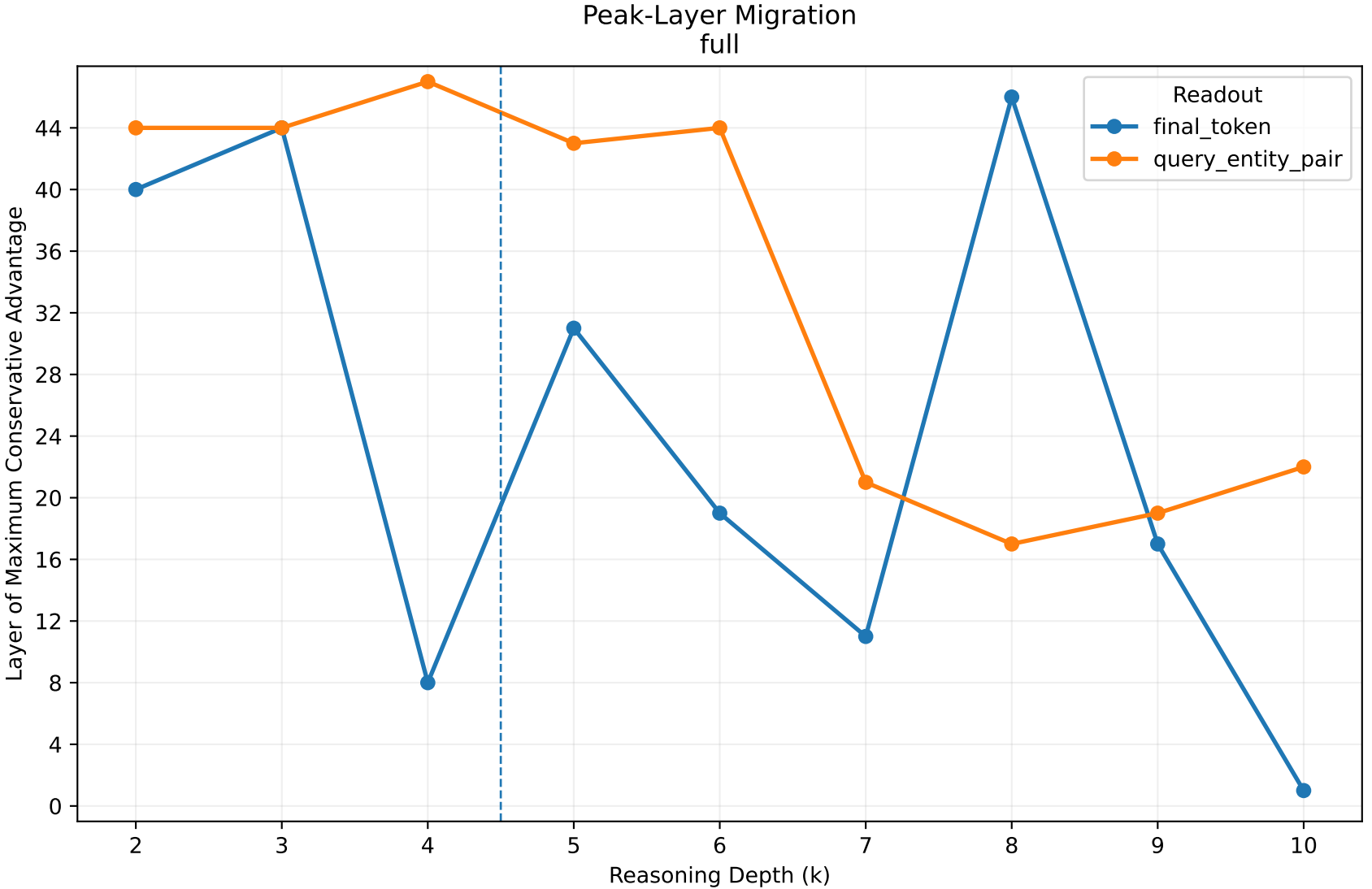

Layer × Composition Depth Map
full | query_entity_pair
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Conservative Advantage

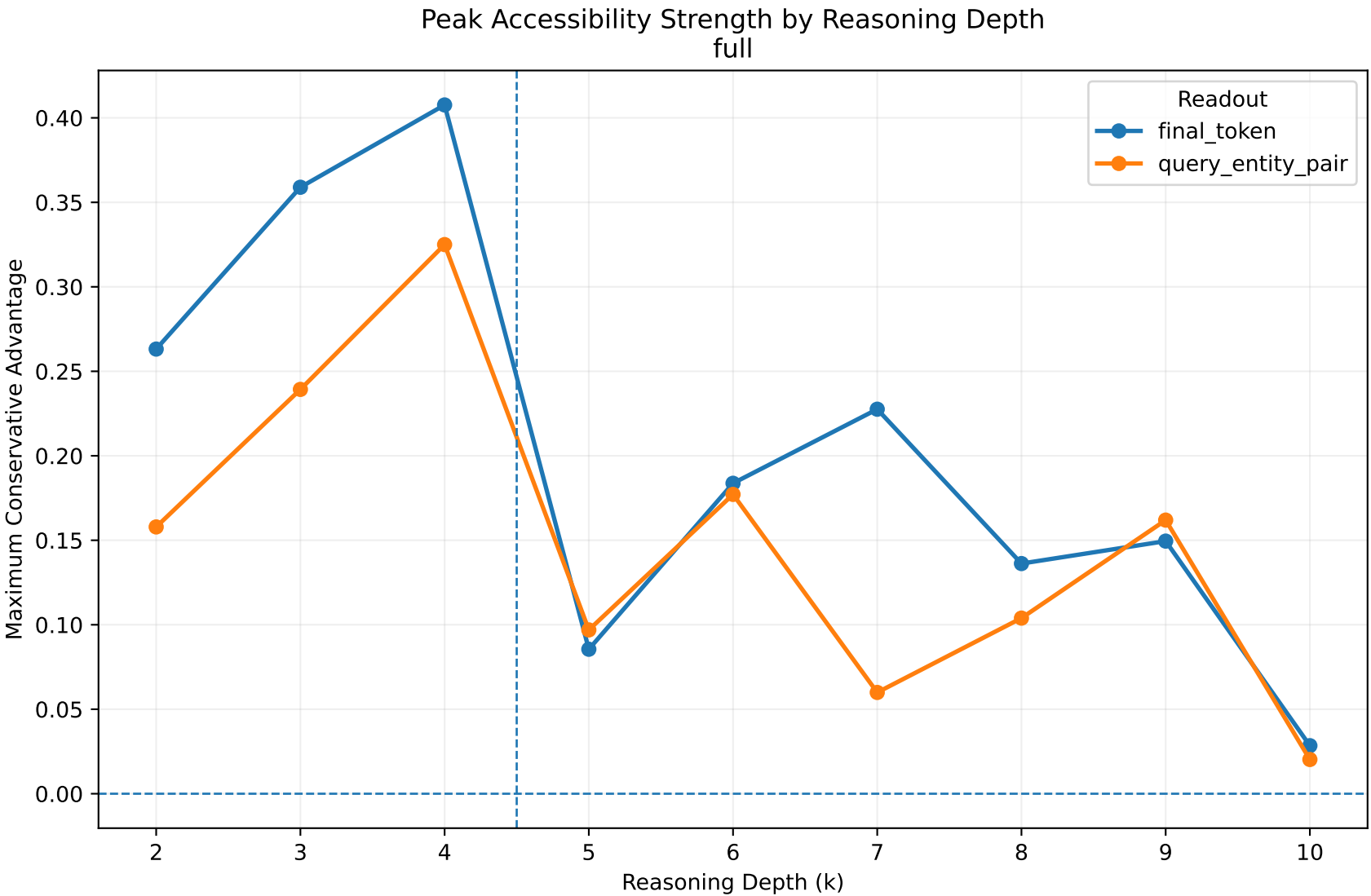

Relational Accessibility Trajectories
full | final_token | seen
Conservative Advantage
Transformer Layer
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
Reasoning Depth
k=2
k=3
k=4

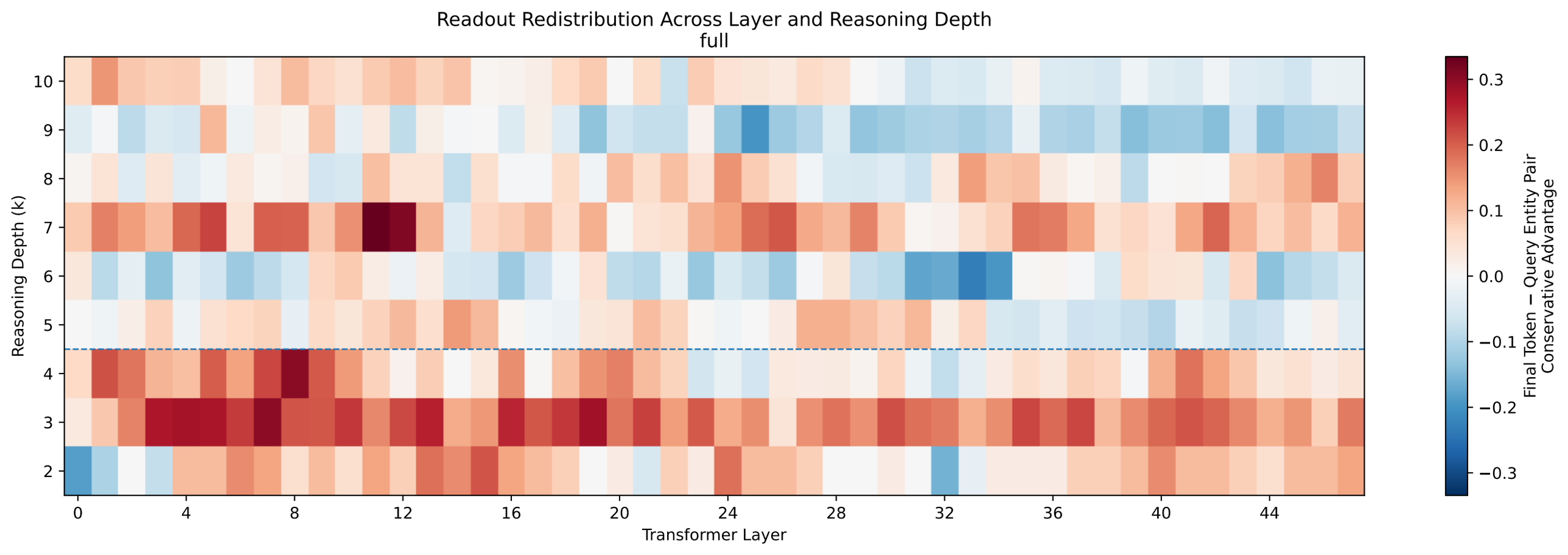

Relational Accessibility Trajectories
full | final_token | unseen
Reasoning Depth
k=5
k=6
k=7
k=8
k=9
k=10
Conservative Advantage
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Transformer Layer

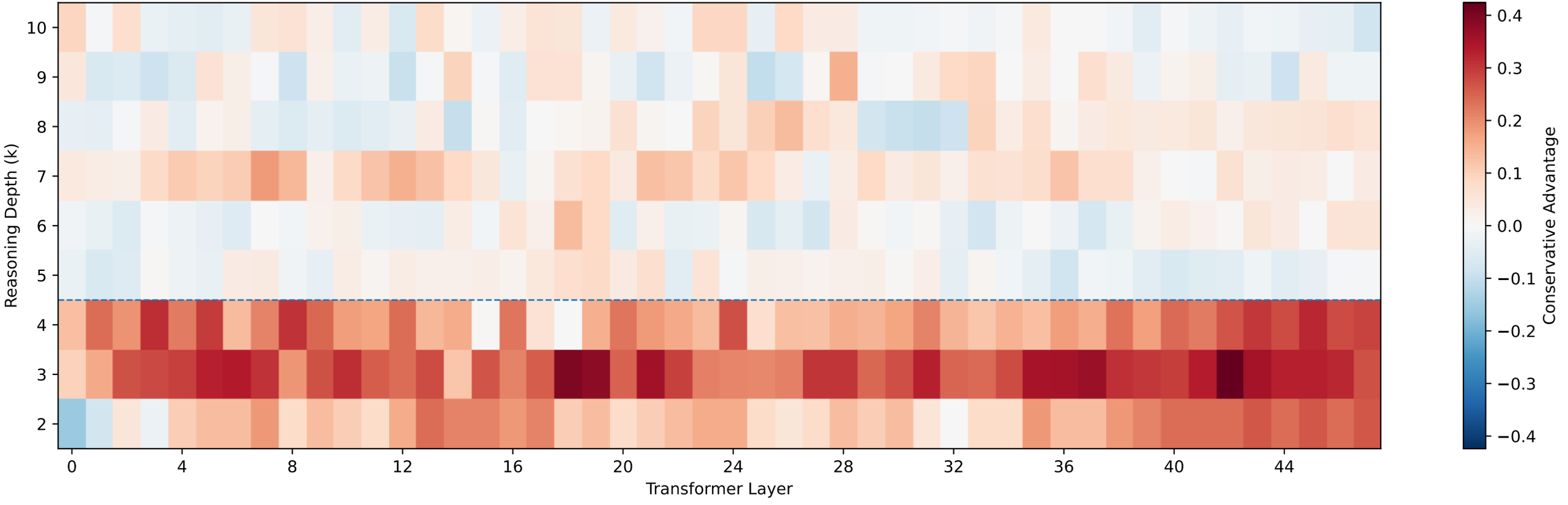

Relational Accessibility Trajectories
full | query_entity_pair | seen
Reasoning Depth
k=2
k=3
k=4
Conservative Advantage
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
Transformer Layer

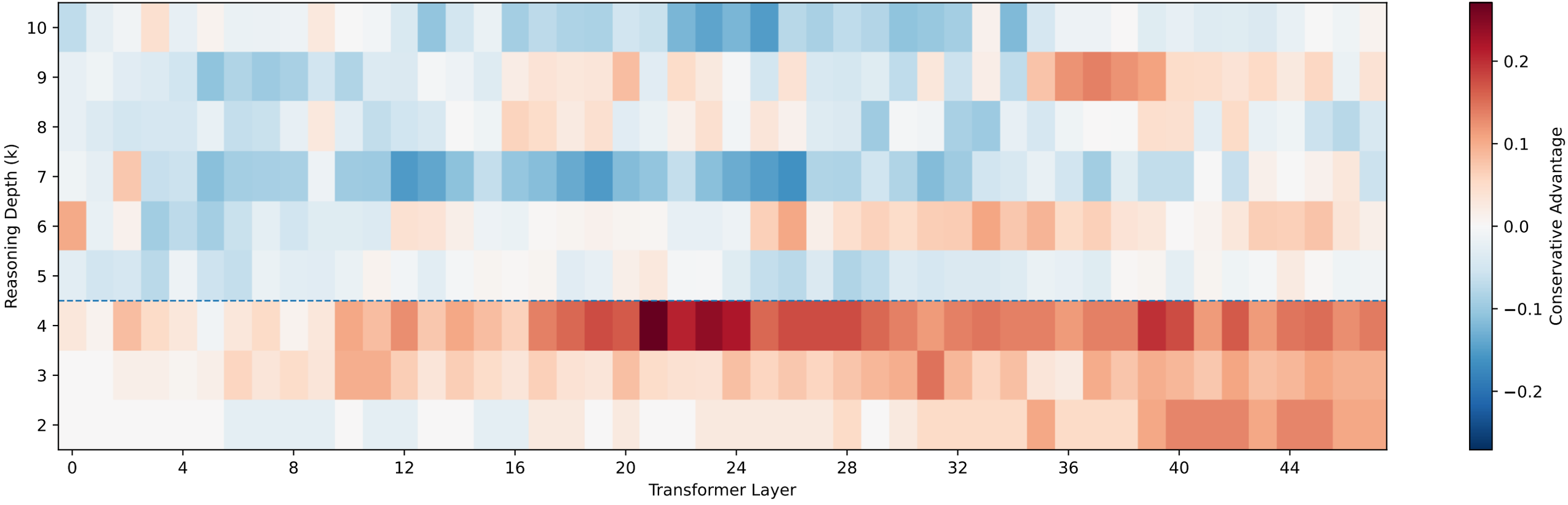

Relational Accessibility Trajectories
full | query_entity_pair | unseen
Reasoning Depth
k=5
k=6
k=7
k=8
k=9
k=10
Conservative Advantage
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
0
10
20
30
40
Transformer Layer

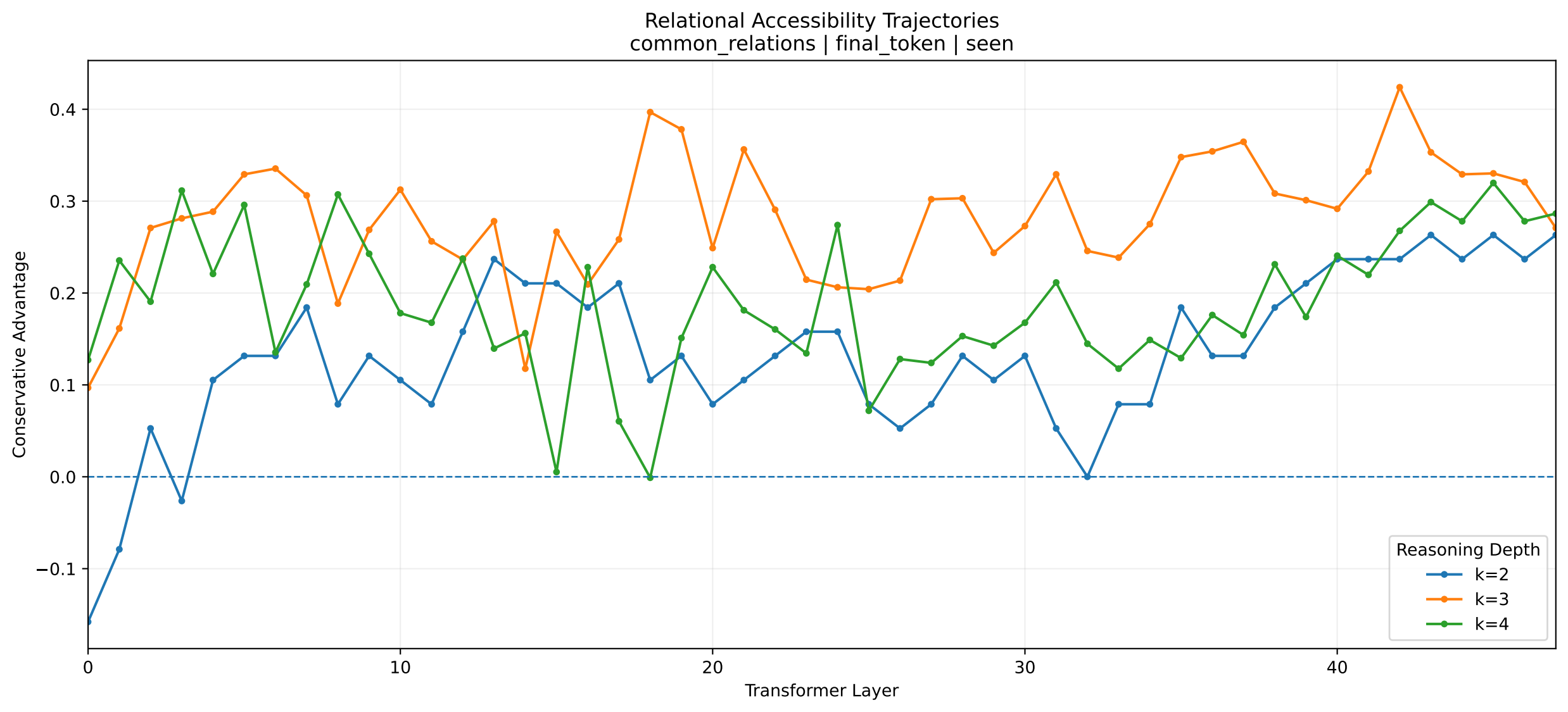

Peak-Layer Migration
full
Readout
final_token
query_entity_pair
Layer of Maximum Conservative Advantage
Reasoning Depth (k)
0
4
8
12
16
20
24
28
32
36
40
44
2
3
4
5
6
7
8
9
10

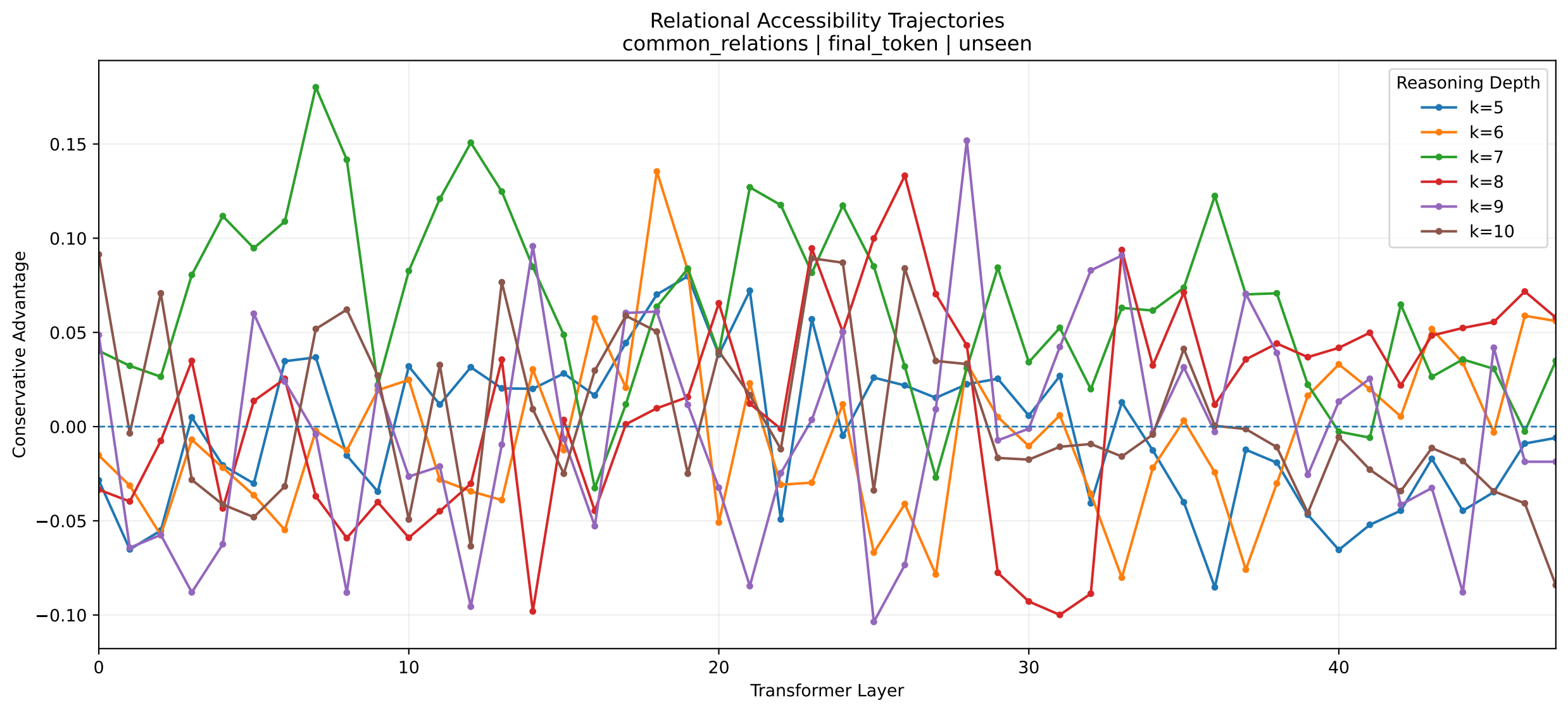

Peak Accessibility Strength by Reasoning Depth
full
Readout
final_token
query_entity_pair
Maximum Conservative Advantage
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
2
3
4
5
6
7
8
9
10
Reasoning Depth (k)

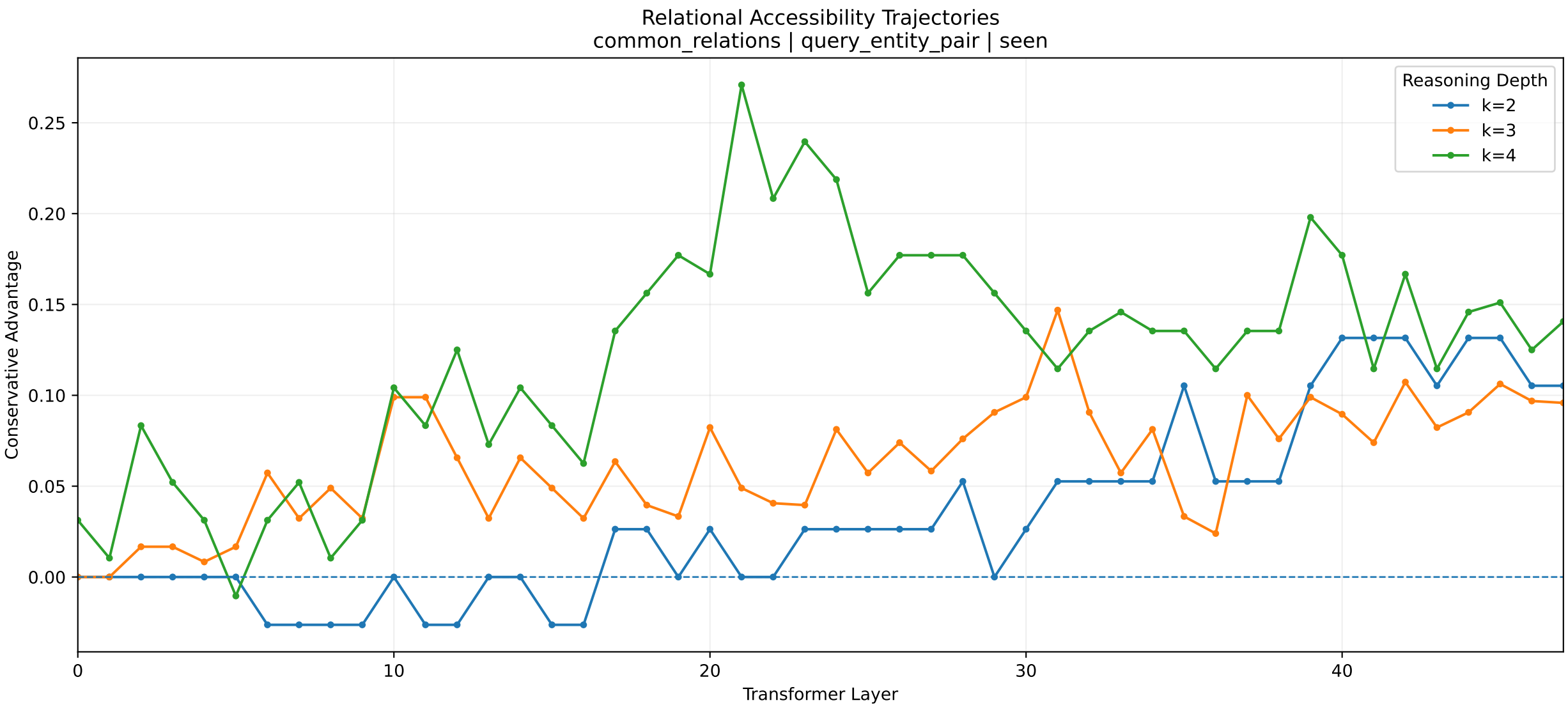

Readout Redistribution Across Layer and Reasoning Depth
full
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Final Token − Query Entity Pair
Conservative Advantage

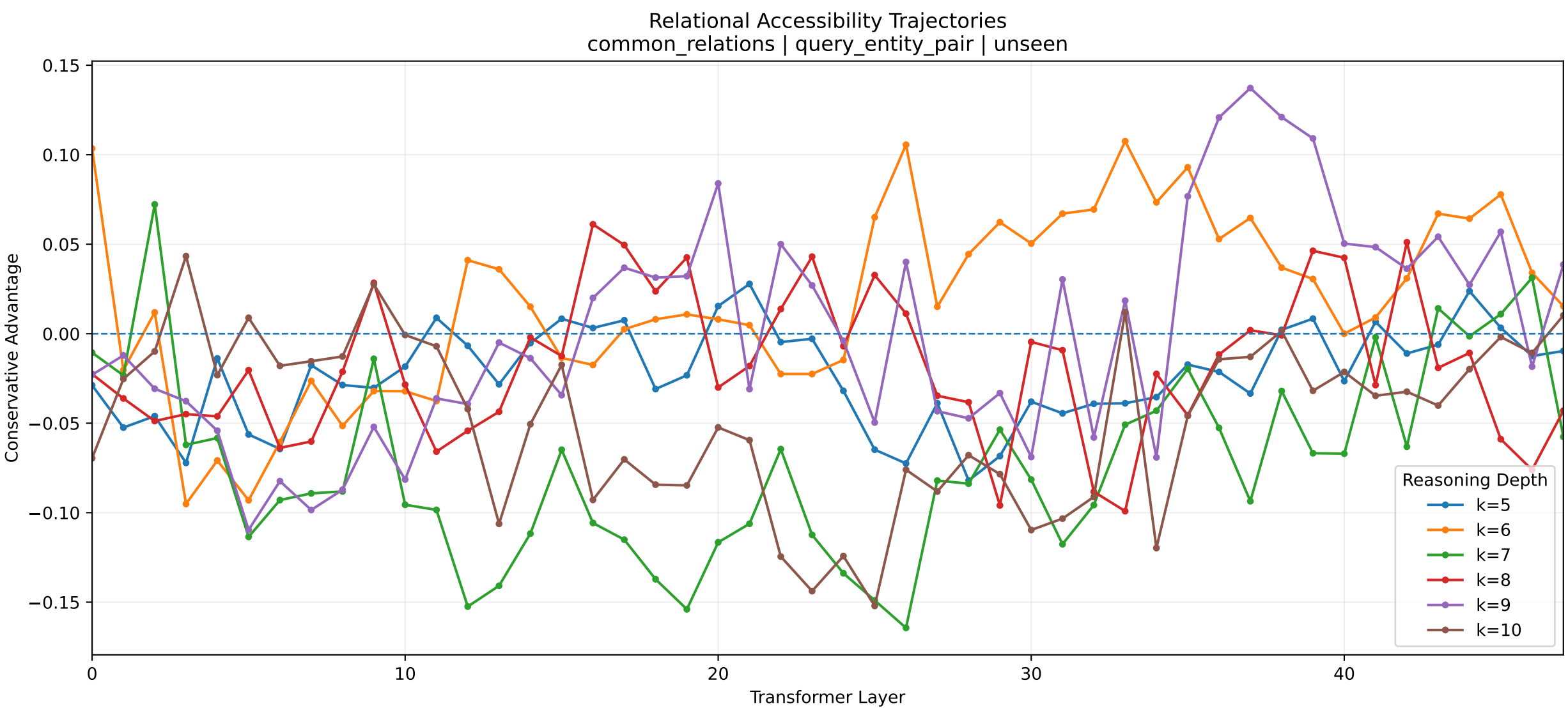

Layer × Composition Depth Map
common_relations | final_token
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Conservative Advantage

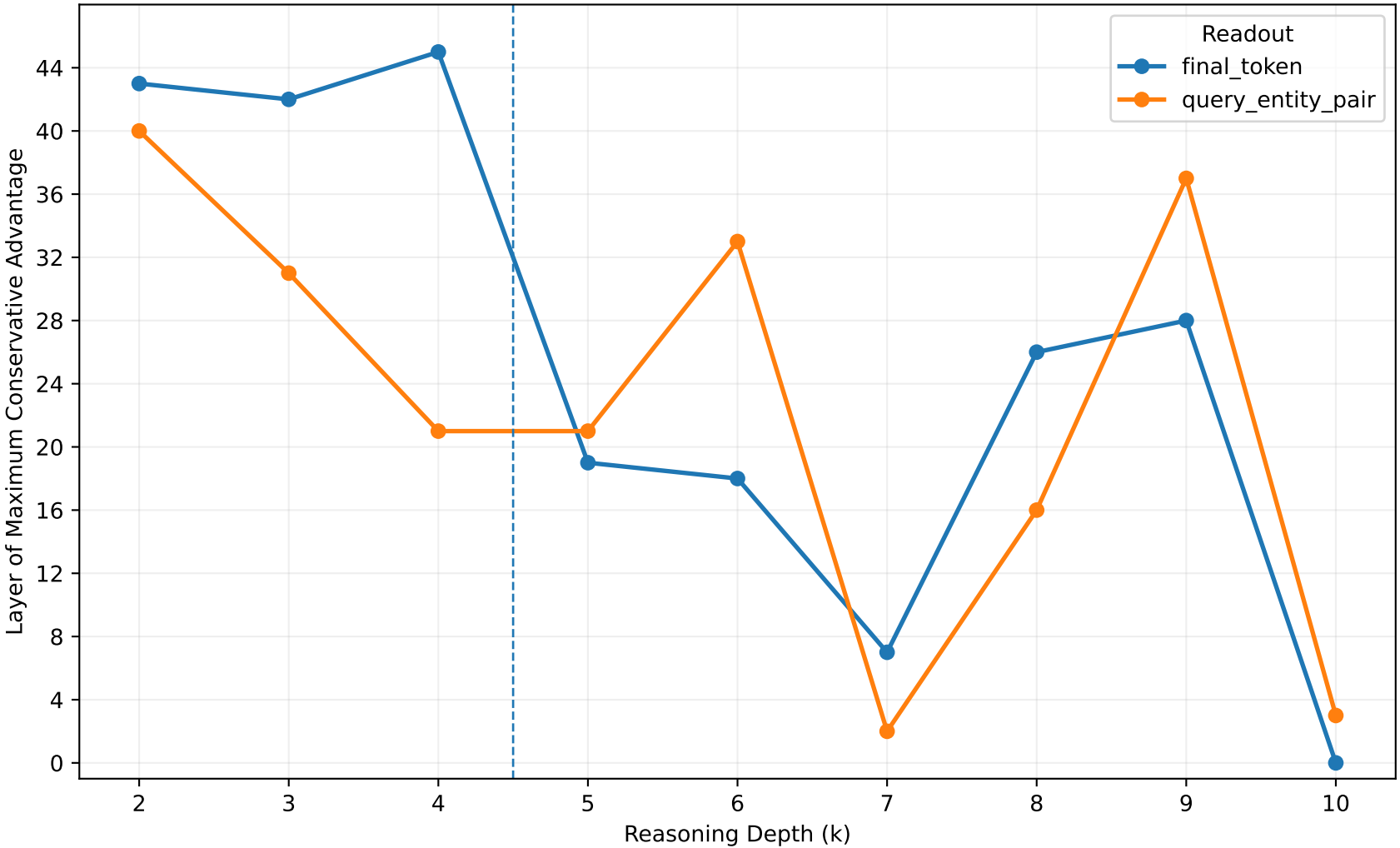

Layer × Composition Depth Map
common_relations | query_entity_pair
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
0.2
0.1
0.0
−0.1
−0.2
Conservative Advantage

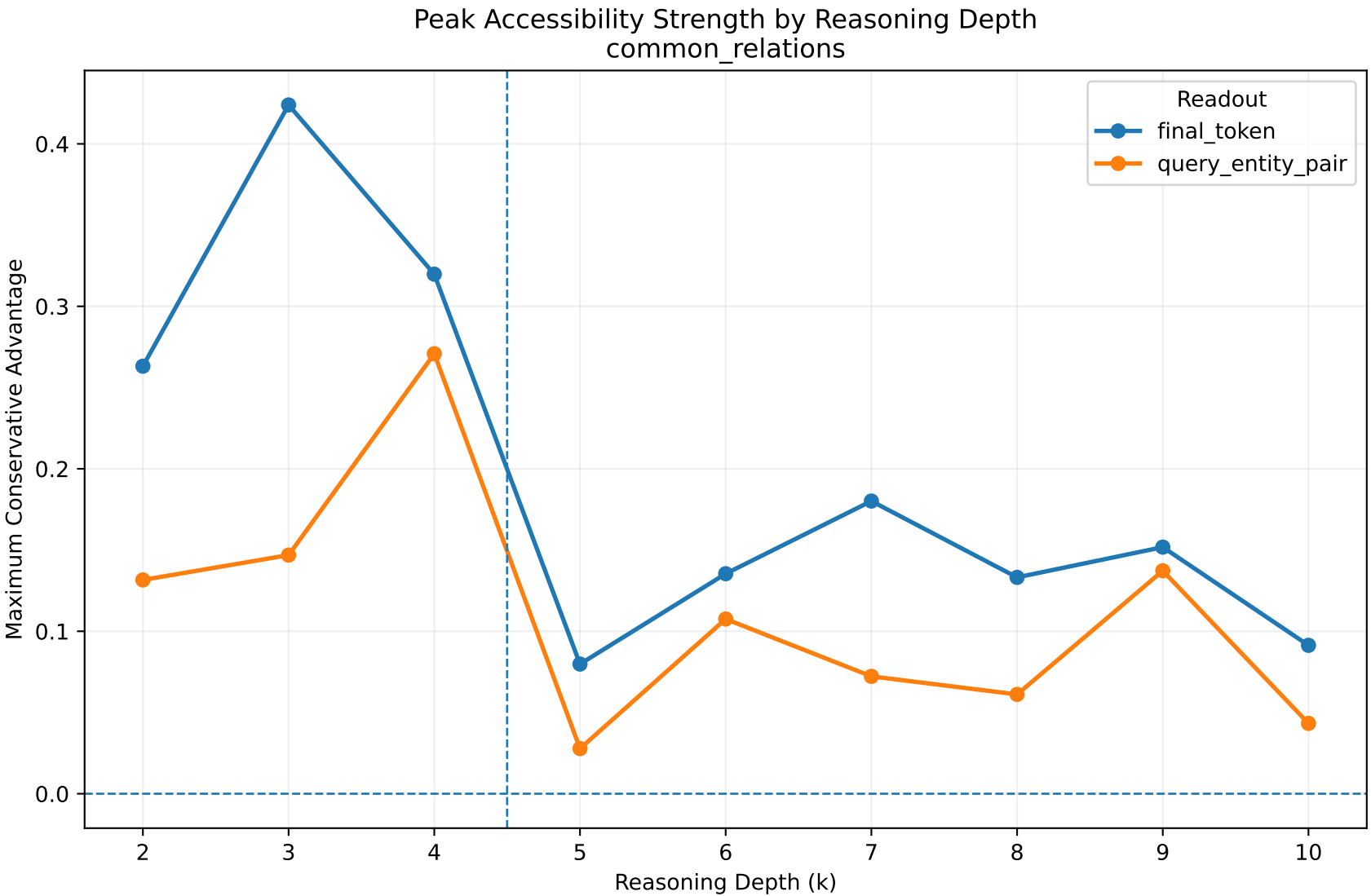

Relational Accessibility Trajectories
common_relations | final_token | seen
Conservative Advantage
Transformer Layer
0.4
0.3
0.2
0.1
0.0
−0.1
0
10
20
30
40
Reasoning Depth
k=2
k=3
k=4

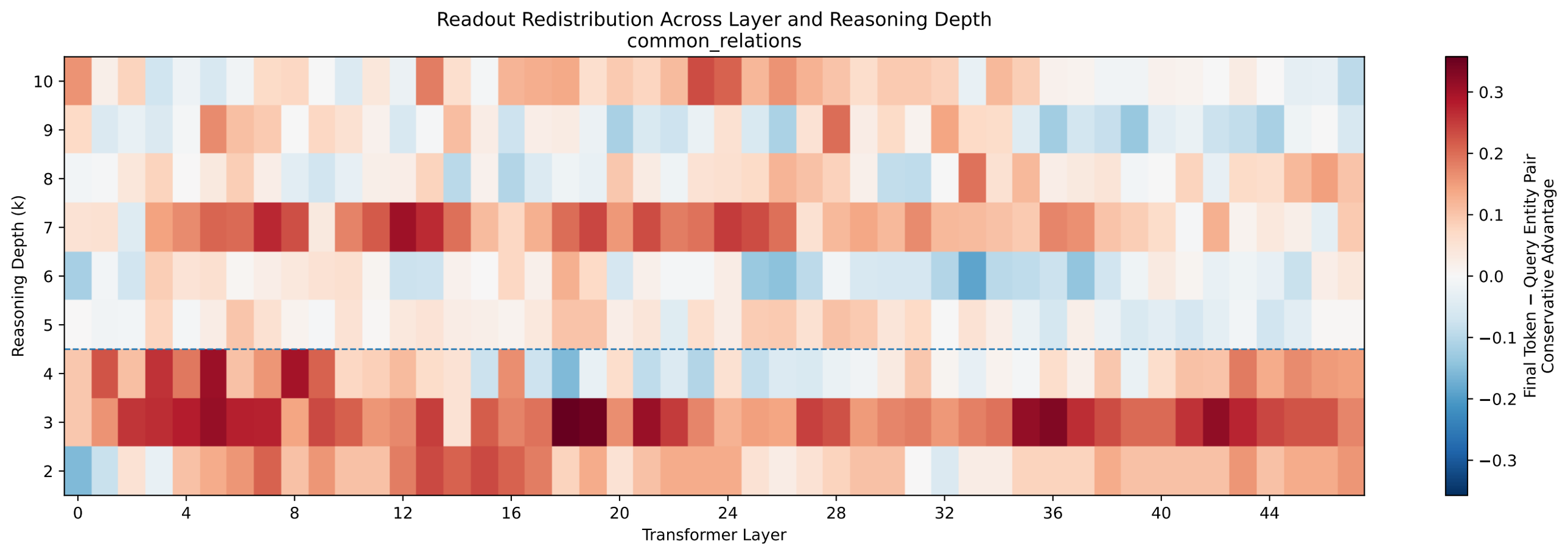

Relational Accessibility Trajectories
common_relations | final_token | unseen
Reasoning Depth
k=5
k=6
k=7
k=8
k=9
k=10
Conservative Advantage
0.15
0.10
0.05
0.00
−0.05
−0.10
0
10
20
30
40
Transformer Layer


Relational Accessibility Trajectories
common_relations | query_entity_pair | seen
Reasoning Depth
k=2
k=3
k=4
Conservative Advantage
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
Transformer Layer


Relational Accessibility Trajectories
common_relations | query_entity_pair | unseen
Conservative Advantage
Transformer Layer
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Reasoning Depth
k=5
k=6
k=7
k=8
k=9
k=10


Peak-Layer Migration
common_relations
Readout
final_token
query_entity_pair
Layer of Maximum Conservative Advantage
Reasoning Depth (k)
0
4
8
12
16
20
24
28
32
36
40
44
2
3
4
5
6
7
8
9
10


Peak Accessibility Strength by Reasoning Depth
common_relations
Readout
final_token
query_entity_pair
Maximum Conservative Advantage
0.0
0.1
0.2
0.3
0.4
2
3
4
5
6
7
8
9
10
Reasoning Depth (k)


Readout Redistribution Across Layer and Reasoning Depth
common_relations
Reasoning Depth (k)
10
9
8
7
6
5
4
3
2
0
4
8
12
16
20
24
28
32
36
40
44
Transformer Layer
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Final Token − Query Entity Pair
Conservative Advantage